

Urban Epidemic Spread Modeling Using Complex Contact Networks and an Artificial Intelligence Surrogate

Jorge Javier Sosa Briseño

Abstract

This article presents a hybrid computational framework for modeling epidemic propagation in an urban population using an aggregated agent-based representation, a modified stochastic block model, spatial friction, hub projection, stratified stochastic SIR dynamics, and an artificial intelligence surrogate. The purpose of the work is to evaluate whether a simplified spatial network can reproduce the qualitative effects observed in detailed agent-based epidemic models while reducing the cost of parameter exploration and calibration.

The model replaces individual agents with aggregated social and non-social population blocks, represents urban activity locations as non-populated hubs, and projects these hubs into effective shortcut edges. Spatial restriction is introduced through a Fermi-Dirac distance function, and epidemic spread is simulated through internal and network-based transmission. Three surrogate architectures are compared: a basic autoencoder, a deep autoencoder with smoothing, and an autoregressive LSTM.

The results show that the LSTM surrogate provides the most stable approximation of the simulator, reaching $R^2 = 0.951$ in the short-distance regime and $R^2 = 0.975$ in the long-distance regime. In the empirical fitting experiment using Saint Petersburg COVID-19 monitoring data from Winter 2022, the LSTM surrogate achieved $R^2 = 0.915$, while the surrogate-assisted SBM validation reached approximately $R^2 = 0.890$. The surrogate reduced the fitting time from approximately 187 s to 31 s, corresponding to an acceleration of about 6×. These results support a hybrid calibration strategy where the surrogate enables fast exploration and the mechanistic simulator provides final validation.

Keywords

epidemic modeling; spatial networks; stochastic block model; hub projection; susceptible–infected–recovered model; artificial intelligence surrogate model; long short-term memory network; parameter calibration; computational epidemiology.

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Introduction

Spatial structure and contact heterogeneity are central factors in epidemic spread. Classical compartmental models are useful for describing aggregate epidemic dynamics, but they usually assume homogeneous mixing and, therefore, cannot explicitly represent how spatial constraints, workplaces, schools, transportation, and

local contact patterns modify the shape of an outbreak. On the other hand, detailed agent-based models can represent individuals and their daily contacts, but their calibration and repeated execution can be computationally expensive.

The reference modeling line for this work is the spatially explicit study of influenza outbreaks in Russian cities, where it was shown that human contact patterns strongly affect the height of the

epidemic peak and the duration of the outbreak. In particular, Leonenko, Arzamastsev, and Bobashev demonstrated that more intensive mixing can produce higher incidence peaks and shorter outbreaks, while more restricted contact structures can flatten and prolong epidemic curves [1]. Related works on synthetic populations for Saint Petersburg also support the importance of explicit spatial and contact allocation rules [2].

The present article proposes an intermediate modeling level between a full individual-based simulator and a purely aggregate compartmental model. Instead of representing every person as a separate agent, the model groups individuals into aggregated population blocks. These blocks are connected using a modified stochastic block model (SBM) with spatial friction and hub projection. The main hypothesis is that this compact representation can preserve the qualitative epidemic mechanisms of interest—local versus long-range mixing, spatial restriction, and peak formation—while making simulation and calibration more efficient.

The purpose of this work is to build and evaluate a hybrid epidemic modeling pipeline that combines: (i) an aggregated ABM representation, (ii) hub-mediated contact projection, (iii) stratified stochastic SIR dynamics, (iv) three AI surrogate architectures (Autoencoders and LSTM), and (v) calibration against real epidemic data. The article focuses on the trade-off between mechanistic fidelity and computational efficiency, demonstrating how the surrogate model can serve as a powerful accelerator for urban epidemiology.

Aggregate Spatial Network Construction

The network is generated as a modified stochastic block model. Each node belongs to one of three categories: social blocks, hubs, or non-social blocks. Social and non-social blocks store resident populations, while hubs represent shared urban activity locations, such as schools, offices, transportation points, and shops.

The block sizes used in the current imple-

mentation are

$$(435, 75, 490),$$

corresponding to social blocks, hubs, and non-social blocks. After filtering to connected populated nodes, the working projected network contains 427 social nodes and 136 non-social nodes (563 nodes total). Since each populated node stores approximately 200–280 individuals, the effective total population is

$$N_{\text{tot}} \approx 563 \times 240 \approx 135,120.$$

The base mixing matrix is defined as

$$\mathbf{P} = \begin{pmatrix} 0.080 & 0.004 & 0.005 \\ 0.004 & 0.000 & 0.001 \\ 0.005 & 0.001 & 0.001 \end{pmatrix}.$$

For two nodes i and j , the base connection probability is

$$p_{\text{base}}(i, j) = B_{z_i, z_j},$$

where z_i and z_j denote the block types of the two nodes.

Each node is assigned a position in an abstract two-dimensional domain of size 38×38 . The distance between two nodes is

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}.$$

Spatial restriction is introduced via a Fermi-Dirac function:

$$f(d_{ij}) = \frac{1}{\exp(\beta_F(d_{ij} - \mu)) + 1}.$$

The parameter μ controls the characteristic distance threshold. Smaller values of μ represent restrictive spatial connectivity, while larger values allow for longer-range mixing.

The probability of link formation depends not only on the block type but also on the contact mechanism. Direct and school-related interactions are distance-sensitive, while office, transport, and shop mechanisms can create longer-range shortcuts:

$$p_{\text{mec}} = \begin{cases} p_{\text{base}} f(d_{ij}), & \text{direct,} \\ p_{\text{base}} f(d_{ij}), & \text{school,} \\ p_{\text{base}}, & \text{office,} \\ p_{\text{base}}, & \text{transport,} \\ p_{\text{base}}, & \text{shop.} \end{cases}$$

If multiple mechanisms are possible, their combined probability by failure theory is

$$p_{\text{final}} = 1 - \prod_m (1 - p_m).$$

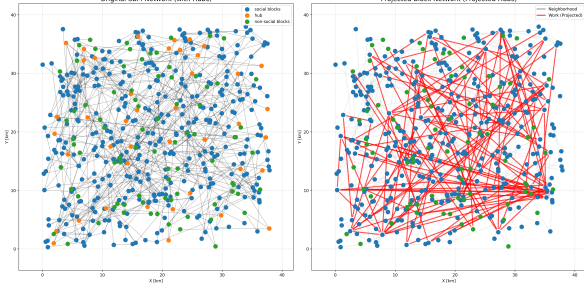


Figure 1: Original and projected aggregate spatial network. The original network contains social blocks, hubs, and non-social blocks. The projected network removes the hubs and replaces them with effective links between blocks.

Hub Projection and Link Weights

Hubs are not retained as final population nodes. Instead, they are projected as effective links between populated blocks. This operation represents the fact that two residential blocks can interact indirectly if they share the same school, workplace, transportation point, or shop.

After projection, the graph may contain multiple links between the same pair of blocks. Before the epidemic simulation, these links are combined into a single weighted graph. If multiple weights $w_{ij}^{(k)}$ exist between nodes i and j , the combined link weight is

$$W_{ij} = 1 - \prod_k (1 - w_{ij}^{(k)}).$$

This expression provides the probability that at least one interaction channel connects the two blocks.

Table 1: Interaction weights used in the aggregated projected network.

Interaction type	Weight
social–social	0.10
social–non-social	0.06
non-social–non-social	0.03
office	0.03
transport	0.02
school	0.99
shop	0.01

Stochastic Epidemic Dynamics

Each populated node stores mobile and static subpopulations:

$$N_i = N_i^M + N_i^E.$$

Social blocks contain between 200 and 280 individuals, with a mobile fraction of 0.7. Non-social blocks contain between 200 and 280 individuals, with a mobile fraction of 0.4. Hubs do not store resident population.

The epidemic process follows a stochastic susceptible–infected–recovered formulation. For each populated node, the model stores

$$S_i^M, I_i^M, R_i^M \quad \text{and} \quad S_i^E, I_i^E, R_i^E,$$

where the superscripts M and E represent mobile and static individuals, respectively.

Before starting the simulation, all individuals are initialized as susceptible:

$$S_i^M(0) = N_i^M, \quad S_i^E(0) = N_i^E,$$

$$I_i^M(0) = I_i^E(0) = R_i^M(0) = R_i^E(0) = 0.$$

The initial number of infected is defined globally as

$$K = \max(K_{\min}, \text{round}(\rho_0 N_{\text{tot}})),$$

where $K_{\min}$ is the minimum number of initial infected, ρ_0 is the initial fraction of infected, and

$$N_{\text{tot}} = \sum_i (N_i^M + N_i^E)$$

is the total population represented in the model.

Initial infected individuals are assigned to nodes using a seeding bias $b_M = 0.7$ towards

the mobile population. In the calibrated configuration, we use $\beta_{\text{household}} = 2.289$, $\beta_{\text{network}} = 0.4469$, and $\delta = 0.9592$, with a simulation horizon of $t_{\text{max}} = 100$ steps.

Initial infected individuals are randomly assigned to nodes according to the number of available susceptibles in each group. For the mobile population, the probability that a mobile initial infection is assigned to node i is

$$P_i^M = \frac{S_i^M}{\sum_j S_j^M}.$$

Similarly, for the static population,

$$P_i^E = \frac{S_i^E}{\sum_j S_j^E}.$$

After this initialization, the corresponding susceptible compartments decrease and the infected compartments increase.

Recovery is sampled using

$$p_{\text{rec}} = 1 - \exp(-\delta).$$

Thus, the number of new recovered individuals is obtained as

$$\Delta R_i^M \sim \text{Binomial}(I_i^M, p_{\text{rec}}),$$

$$\Delta R_i^E \sim \text{Binomial}(I_i^E, p_{\text{rec}}).$$

Internal transmission within a node is determined by the fraction of infected within the local population unit:

$$\lambda_i^{\text{int}} = \beta_{\text{household}} \frac{I_i^M + I_i^E}{N_i},$$

$$p_i^{\text{int}} = 1 - \exp(-\lambda_i^{\text{int}}).$$

New internal infections are sampled from the mobile and static susceptible populations:

$$\Delta I_i^{M,\text{int}} \sim \text{Binomial}(S_i^M, p_i^{\text{int}}),$$

$$\Delta I_i^{E,\text{int}} \sim \text{Binomial}(S_i^E, p_i^{\text{int}}).$$

External transmission through the network is determined by mobile infected individuals in neighboring nodes:

$$X_i = \sum_{j \in \mathcal{N}(i)} W_{ij} \frac{I_j^M}{N_j^M}.$$

The external force of infection is defined as

$$\lambda_i^{\text{ext}} = \beta_{\text{network}} X_i,$$

with infection probability

$$p_i^{\text{ext}} = 1 - \exp(-\lambda_i^{\text{ext}}).$$

External infection applies only to mobile susceptible individuals:

$$\Delta I_i^{M,\text{ext}} \sim \text{Binomial}(S_i^M, p_i^{\text{ext}}).$$

Therefore, the model separates two transmission mechanisms. Internal transmission affects both mobile and static individuals within the same population unit, while external transmission acts through the projected network of contacts and only affects mobile susceptible individuals. This structure allows representing both local contagion within population blocks and mobility-driven contagion between different population units.

Artificial Intelligence Surrogate Model

The artificial intelligence surrogate model is designed to approximate the stochastic mapping from epidemic and spatial parameters to infected trajectories. Three architectures were evaluated:

1. **Basic Autoencoder:** Directly reconstructs the epidemic curve from parameters through a 16-dimensional latent space. It is fast but lacks explicit temporal memory.
2. **Deep Autoencoder with Smoothing:** Uses a deeper encoder-decoder structure and incorporates shape penalties (peak, slope, and curvature) to reduce instability.
3. **Autoregressive LSTM:** Parameters initialize a 2-layer LSTM that generates the trajectory sequentially ($I(t-1) \rightarrow I(t)$). This architecture best captures the temporal dependencies of the epidemic process.

The surrogate models are trained with synthetic data produced by the SBM-SIR simulator. This hybrid approach enables fast parameter exploration while preserving mechanistic interpretability.

Calibration Protocol and Evaluation

Calibration uses two stages. First, the surrogate model quickly searches for a good parameter configuration. Second, the original simulator is initialized from the surrogate model’s solution and refined through repeated stochastic simulations.

The adjusted parameter vector is

$$\theta = (\beta_{\text{network}}, \beta_{\text{household}}, \delta, \mu, \tau),$$

where τ is a temporal shift. The model output is shifted in time and compared with the normalized empirical curve:

$$L = \text{MSE} \left(\frac{C_{\text{real}}(t)}{\max C_{\text{real}}}, \frac{\hat{I}(t - \tau)}{\max \hat{I}} \right).$$

After obtaining the best shape, a scale coefficient is calculated. The predicted reported cases are then

$$\hat{C}(t) = s\hat{I}(t).$$

Thus, the fit to real data is phenomenological: the model reproduces the shape of the epidemic curve and then rescales the normalized fraction of infected to reported cases.

Results and Discussion

The spatial experiment compares two regimes. The restrictive regime uses a smaller distance threshold and produces a flatter outbreak. The freer regime uses a larger distance threshold and produces a more pronounced outbreak. Table 2 summarizes the final quantitative comparison.

Table 2: Comparison between restrictive and freer spatial regimes.

Metric	Restrictive	Freer
Peak	1241 $\pm$ 282	7523 $\pm$ 331
Cumulative	35380 $\pm$ 6605	76344 $\pm$ 1284
Final I	244 $\pm$ 194	0 $\pm$ 0

The results confirm the expected qualitative mechanism. A stronger spatial restriction slows down the epidemic and reduces the peak,

while a freer mixing increases the effective connectivity and concentrates infections into a higher and earlier peak.

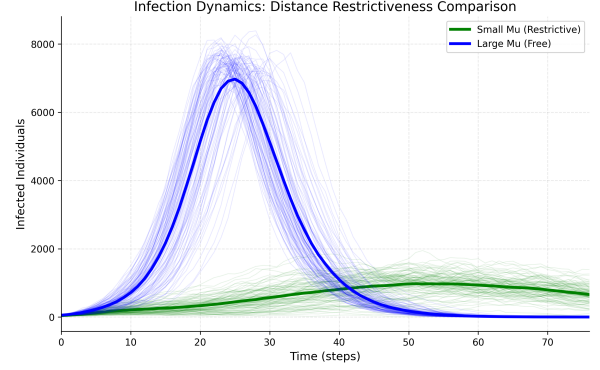


Figure 2: Infection dynamics under restrictive and freer spatial regimes. A freer spatial mixing produces a substantially higher and earlier infection peak.

The surrogate comparison shows that the LSTM is the most reliable architecture for both short and long distance scenarios. In the short-distance regime ($\mu = 5.0$), the epidemic is more local and noisy, and the LSTM follows the simulator curve with $R^2 = 0.951$. In the long-distance regime ($\mu = 15.0$), the epidemic spreads more freely and the LSTM reaches $R^2 = 0.975$.

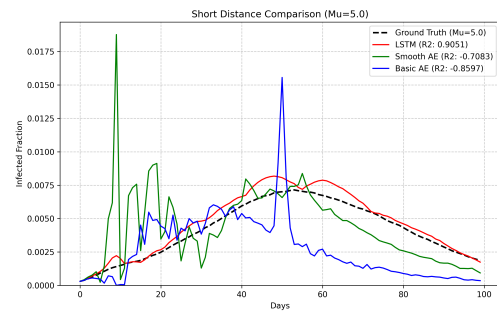


Figure 3: Surrogate comparison in the short-distance regime.

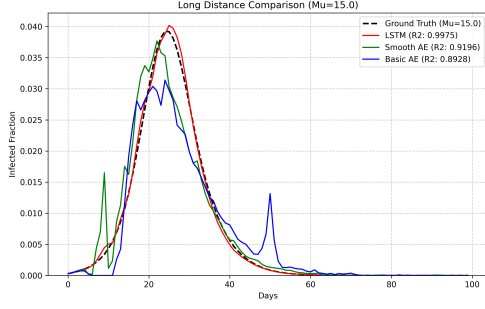


Figure 4: Surrogate comparison in the long-distance regime.

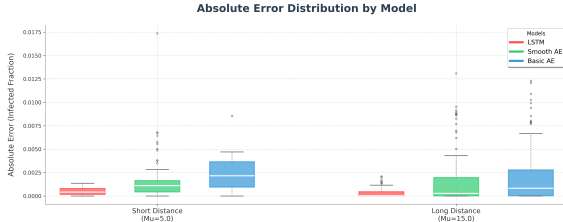


Figure 5: Error distribution of surrogate models in short and long distance regimes. The LSTM shows the smallest and most stable error distribution.

The speed comparison also confirms the value of the surrogate. Autoencoder inference takes approximately 0.03 s per curve, LSTM inference takes approximately 0.05 s per curve, while one SBM simulation takes approximately 3.5 s per curve.

The final fitting experiment uses COVID-19 monitoring data from Saint Petersburg for Winter 2022. Both the LSTM surrogate and the surrogate-assisted SBM model reproduce the main epidemic peak.

Table 3: Real-data fitting performance.

Model	R^2	Fitting time
LSTM surrogate	0.915	~ 31 s
Surrogate + SBM	≈ 0.890	~ 187 s

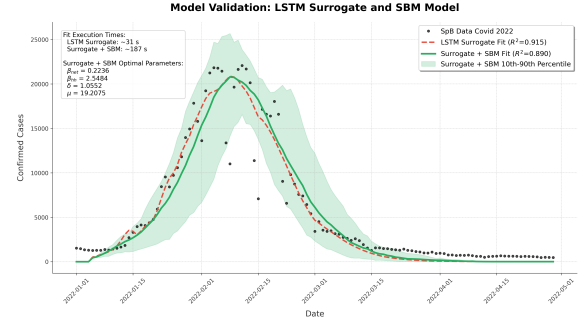


Figure 6: Empirical fitting of the surrogate and SBM model to Saint Petersburg COVID-19 monitoring data, Winter 2022.

The surrogate-assisted fitting is approximately $6\times$ faster than the simulator-based validation. This result supports the proposed hybrid strategy.

The results demonstrate a clear trade-off between mechanistic fidelity and computational efficiency. The original SBM simulator obtains a slightly better mechanistic fit, but the surrogate substantially reduces calibration time. For this reason, the combined use of both models is methodologically convenient: the surrogate allows for a rapid parameter search and the original simulator allows for a final validation truer to the epidemic mechanism of the network.

Conclusion

The proposed framework provides a compact and computationally efficient approach for studying epidemic spread in urban populations with spatial structure. The model is not a full individual-based reconstruction of a city. Instead, it is an optimized aggregate model that retains the essential role of spatial contact structure, hub-mediated interaction, and mobile population groups.

The main finding is that the model reproduces the expected qualitative effect of spatial mixing. Restrictive contact structures generate flatter and more prolonged outbreaks, while freer contact structures generate sharper and higher epidemic peaks. This result is consistent with the spatial epidemic modeling literature and supports the use of aggregate network models as interpretable proof-of-concept tools.

The artificial intelligence surrogate model successfully approximates the epidemic curves

generated by the original simulator and allows for faster parameter exploration. The final comparison shows that the original simulator provides slightly better fitting performance, while the surrogate model offers a clear computational advantage. Therefore, the most practical strategy is not to choose one model over the other but to combine them: the surrogate model can be used for rapid calibration, and the original simulator can be employed for final mechanistic refinement.

Future work can leverage the modular potential of the proposed framework to expand its scope. In particular, the neural network could incorporate new input parameters related to network structure, mobility, and initial conditions, allowing for the representation of more diverse epidemic scenarios. Likewise, applying the model to other reported epidemiological curves would allow for evaluating its robustness in different contexts. Finally, analyzing its predictive capacity would strengthen its utility as a tool for rapid exploration, calibration, and support for the study of future scenarios.

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